WORKED SOLUTION

STEP III 1987, Question 9

Pure • Direct Products of Groups

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s3-q9

The key idea

Everything in $G \times H$ happens one coordinate at a time, so each group axiom and each property of $G \times H$ comes down to the same property in G and in H separately. For the true/false parts, a "true" needs a general proof, while a "false" needs just one solid obstruction. Isomorphic groups agree in everything that can be described using only the multiplication, such as their size and the orders of their elements. So we compare those.

Solution

Throughout, e_G and e_H are the identities of G and H, and we write g^{-1} and h^{-1} for inverses in G and H.

$G \times H$ is a group

We check the four axioms.

Closure. If $g_1, g_2 \in G$ and $h_1, h_2 \in H$, then $g_1 * g_2 \in G$ and $h_1 \circ h_2 \in H$, because G and H are closed. So $(g_1, h_1)(g_2, h_2) = (g_1 * g_2,\ h_1 \circ h_2) \in G \times H$.

Associativity. For any three elements,

$$\begin{aligned}((g_1, h_1)(g_2, h_2))(g_3, h_3) &= ((g_1 * g_2) * g_3,\ (h_1 \circ h_2) \circ h_3) \\ &= (g_1 * (g_2 * g_3),\ h_1 \circ (h_2 \circ h_3)) \\ &= (g_1, h_1)((g_2, h_2)(g_3, h_3)),\end{aligned}$$

where the middle step uses associativity in G and in H.

Identity. The pair (e_G, e_H) satisfies

$$(e_G, e_H)(g, h) = (e_G * g,\ e_H \circ h) = (g, h),$$

and similarly $(g, h)(e_G, e_H) = (g, h)$. So it is an identity.

Inverses. Given (g,h), the pair (g^{-1},h^{-1}) lies in $G\times H$, and

$$(g,h)(g^{-1},h^{-1}) = (g*g^{-1},\ h\circ h^{-1}) = (e_G,e_H),$$

and the product the other way round is also (e_G,e_H). So every element has an inverse. All four axioms hold, so $G\times H$ is a group under this multiplication.

Part (i): true

If G and H are abelian, then so is $G\times H$. For any two elements,

$$(g_1,h_1)(g_2,h_2) = (g_1*g_2,\ h_1\circ h_2) = (g_2*g_1,\ h_2\circ h_1) = (g_2,h_2)(g_1,h_1),$$

using commutativity in G and in H.

If $G\times H$ is abelian, then so are G and H. Take any $g_1,g_2\in G$. Since $G\times H$ is abelian,

$$(g_1*g_2,\ e_H) = (g_1,e_H)(g_2,e_H) = (g_2,e_H)(g_1,e_H) = (g_2*g_1,\ e_H).$$

Ordered pairs are equal only when their first coordinates are equal, so $g_1*g_2 = g_2*g_1$, and G is abelian. Using pairs (e_G,h_1) and (e_G,h_2) in the same way shows that H is abelian.

> **Ask yourself** Why do we need the identity e_H in the second direction?

We need some second coordinate to make elements of $G\times H$ from g_1 and g_2. Any fixed $h\in H$ would do, since only the first coordinates matter in the comparison. A group always contains an identity, so e_H is a safe choice that we know exists.

Part (ii): true

Let $\widetilde{G} = \{(g,e_H) : g\in G\}$, and define $\phi\colon G\to\widetilde{G}$ by $\phi(g) = (g,e_H)$.

$\widetilde{G}$ is a subgroup of $G\times H$. It contains the identity (e_G,e_H). It is closed under products, since $(g_1,e_H)(g_2,e_H) = (g_1*g_2,e_H)$. It is closed under inverses, since $(g,e_H)^{-1} = (g^{-1},e_H)$.

ϕ is an isomorphism. It is onto $\widetilde{G}$ by definition. It is one-to-one, because $(g_1,e_H) = (g_2,e_H)$ forces $g_1 = g_2$. It preserves the operation, because

$$\phi(g_1*g_2) = (g_1*g_2,\ e_H) = (g_1,e_H)(g_2,e_H) = \phi(g_1)\phi(g_2).$$

So $G\times H$ contains the subgroup $\widetilde{G}$, which is isomorphic to G.

Part (iii): false

Write both groups additively: $\mathbb{Z}_n = \{0,1,\ldots,n-1\}$ under addition mod n. In $\mathbb{Z}_2\times\mathbb{Z}_2$, every element $x=(a,b)$ satisfies

$$x+x = (2a,2b) = (0,0),$$

so every element has order 1 or 2. In $\mathbb{Z}_4$, by contrast, $1+1=2\neq 0$, so 1 has order 4.

Suppose there were an isomorphism $\phi\colon \mathbb{Z}_4 \to \mathbb{Z}_2 \times \mathbb{Z}_2$. Then

$$\phi(2) = \phi(1+1) = \phi(1) + \phi(1) = (0,0) = \phi(0),$$

because $\phi(1)+\phi(1)=(0,0)$ by the calculation above. The last equality holds because an isomorphism sends the identity to the identity. This gives $\phi(2)=\phi(0)$ with $2\neq 0$, so ϕ is not one-to-one, which is a contradiction. Hence $\mathbb{Z}_2 \times \mathbb{Z}_2$ is *not* isomorphic to $\mathbb{Z}_4$. (In short, $\mathbb{Z}_4$ is cyclic and $\mathbb{Z}_2 \times \mathbb{Z}_2$ is not.)

Part (iv): false

An isomorphism is a bijection, so isomorphic groups have the same number of elements. Now $|S_n| = n!$, because there are $n!$ arrangements of n objects, and $|G \times H| = |G|\,|H|$, since each pair is a free choice of g and of h. So

$$|S_2 \times S_3| = 2! \times 3! = 12, \qquad |S_6| = 6! = 720.$$

These sizes differ, so $S_2 \times S_3$ is *not* isomorphic to S_6.

Checks

- By brute force on small examples (such as $\mathbb{Z}_2 \times \mathbb{Z}_3$, $\mathbb{Z}_2 \times S_3$, $S_3 \times S_3$ and $S_2 \times S_3$), the product satisfies all four axioms with identity (e_G, e_H). It is abelian exactly when both factors are, and $g \mapsto (g, e_H)$ is an injective homomorphism.
- The element orders in $\mathbb{Z}_2 \times \mathbb{Z}_2$ are $1,2,2,2$, and in $\mathbb{Z}_4$ they are $1,2,4,4$, so the two groups really are different.
- Listing $S_2 \times S_3$ gives 12 elements, against 720 in S_6.

Common slips

- Proving only one direction of the "if and only if" in (i). The converse is the part that needs the trick of putting e_H in the second coordinate.
- In (ii), saying "G is a subgroup of $G \times H$". Strictly, G is not even a subset of $G \times H$. The subgroup is $G \times \{e_H\}$, which is *isomorphic* to G.
- In (iii), noting that both groups have order 4 and stopping there. Equal size is necessary for an isomorphism but not sufficient, so we need a property that tells the groups apart.

Beyond STEP

This is the *direct product* construction, the simplest way of building new groups from old. A major result of a first course is that every finite abelian group is a direct product

of cyclic groups. A related fact is that $\mathbb{Z}_m \times \mathbb{Z}_n \cong \mathbb{Z}_{mn}$ exactly when $\gcd(m, n) = 1$, which is the group-theoretic face of the Chinese remainder theorem. Part (iii) is the smallest case where this fails. For more, see Allenby, *Rings, Fields and Groups*.

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